Igor Babin

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Education

Ukrainian Catholic University, Computer Science (Data Science) — Lviv, Ukraine, 2022 – 2024
- Thesis topic: "Image inpainting through refinement of features space"

National University of Kyiv-Mohyla Academy, Applied Mathematics — Kyiv, Ukraine, 2017 – 2021

Experience

NDA (military, DefTech), Senior Machine Learning / Computer Vision Engineer — Kyiv, Ukraine, June 2025 – present

End-to-end computer vision solutions — from data collection and annotation preparation to model training, experimentation, and deployment on embedded devices with limited computational resources.
- Proficient in Python, PyTorch, and MLflow for model development and optimization workflows
- Real-time deployment on hardware such as Raspberry Pi 5, Pi Zero, and Hailo NPUs
- Object detection, tracking, and visual navigation tasks
- Model compression techniques (distillation, pruning, quantization) to achieve real-time performance while maintaining robust accuracy

Ukrainian Catholic University / SET University, Lecturer of Generative AI course (Computer Vision part) / Efficient Machine Learning — Lviv / Kyiv, Ukraine, Sept 2024 – present

Preparing and delivering lectures on Generative Computer Vision for Master's students at UCU, and lecturing on Efficient Machine Learning at SET University.
- Designing and teaching the Generative AI course (for M.Sc. students) — Computer Vision part
- Mentoring master's theses, helping future ML engineers grow their expertise

ADVA Soft, Machine Learning Team Lead — Lviv, Ukraine, June 2023 – June 2025

Leads a team of machine learning engineers in developing and implementing machine learning models and systems. Provides technical leadership, manages project timelines, and ensures the successful delivery of ML-powered solutions on devices through the CoreML framework.
- Lead a team of machine learning engineers, providing technical guidance and support
- Collaborate with other teams to help integrate ML solutions into projects
- Collaborate with product owners to define project requirements and goals
- Design and implement machine learning models and algorithms to evaluate and validate ML models' performance and accuracy
- Stay updated with the latest ML algorithms, frameworks, and techniques advancements

ADVA Soft, Machine Learning Engineer | Computer Vision — Lviv, Ukraine, Mar 2021 – June 2023

Develop and implement machine learning models. Work closely with software engineers to build and deploy ML-powered solutions that address business challenges and enhance product functionality.
- Develop and implement machine learning models and algorithms
- Integrate ML models into software systems and applications
- Optimize and fine-tune ML models for improved performance and accuracy

AI Meetups, Lecturer — Lviv, Ukraine, Oct 2023 – June 2024
- Delivering lectures on various AI-related topics

- Searching for speakers in different domains
- Managing technical materials of other speakers

mentor.sh, Mentor and Consultant — Kyiv, Ukraine
July 2025 – present

Binary Studio, Software Engineer — Kyiv, Ukraine
June 2019 – Sept 2020

Volunteer

Community meetups and hackathons, Speaker — Ukraine
Jan 2024 – present
Seasonally delivering lectures on Computer Vision, Machine Learning, and AI topics at various meetups and hackathons, including Ukrainian Catholic University, AI House, AI meetups, and Kyiv AI Day.

Conferences, Conference Reviewer — Jan 2026 – present
Serve as a conference reviewer evaluating paper quality for CVPR 2026 and ECCV 2026 submissions.

Awards

IYMC Gold Honour
Awarded by the International Youth Math Challenge.

Skills

Machine Learning / Computer Vision

Embedded AI

Frameworks & Tools

Leadership

Languages

Ukrainian
Native or bilingual

English
Full professional proficiency

Certificates

TensorFlow: Advanced Techniques

MLOps Mini-course

TensorFlow Developer Certificate

Deep Learning Specialization

Strategic Leadership

Publications

DeVOS: Flow-Guided Deformable Transformer for Video Object Segmentation
Igor Babin

Global Motion Understanding in Large-Scale Video Object Segmentation
Igor Babin

Interests

Computer Vision

Embedded AI